

DIEGO ZIDANE REYES GOMEZ

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Technical Skills

Programming Languages: Python, C++, Java, JavaScript, HTML/CSS, R, Kotlin

AI/ML: PyTorch, RAG, LLM Agents, Prompt Engineering, Vector Search, Model Evaluation

Frameworks: React, Next.js, Node.js, Express.js, FastAPI, Tailwind CSS

Cloud/DevOps: AWS, S3, Docker, Linux, Git, REST APIs, MCP, Postman

Developer Tools: VS Code, Cursor, Unreal Engine 5, Android Studio, Figma, Adobe Creative Cloud

Professional Experience

Werfen

June 2026 – September 2026

AI Engineer Intern (Marketing & Digital Innovation Team)

San Diego, CA

- Developed a RAG-based Marketing Agent within WALT, Werfen's multi-agent AI platform, to automate research and response generation for high-volume technical product inquiries from global affiliates, streamlining a manual workflow that often required multiple days to resolve.
- Built document-grounded retrieval, intent classification, clarification, and escalation workflows across 5 product lines, generating traceable responses from validated technical documentation while preventing unsupported answers.
- Engineered document-family matching and routing using normalized SAP metadata, document identifiers, language attributes, revisions, and file variants to improve retrieval accuracy and separate unrelated technical documents.
- Developed and tested AI tooling including multi-product retrieval, multilingual query handling, automated email generation, and AWS S3-based part-number image retrieval; validated model behavior, API payloads, filters, and fallback logic through regression testing and Postman.
- Collaborated with Marketing and Digital Innovation stakeholders to translate business requirements into production-ready AI workflows, configure instrument-specific knowledge connectors, and prepare the agent for validation and deployment.

San Diego Supercomputer Center (SDSC)

June 2023 – June 2026

Software Engineer Intern

San Diego, CA

- Redesigned *Workload*, a full-stack web management application for SDSC's Research Data Services department, using Next.js, React, TypeScript, and Tailwind CSS to deliver a responsive Kanban interface for monitoring team assignments across GitLab, Zendesk, and Trello.
- Implemented member selection, project specific views, issue filtering, and five status-based columns to organize active, blocked, pending, completed, and uncategorized work while excluding closed items.
- Containerized the frontend, Go backend, and PostgreSQL database using Docker Compose and configured API endpoints and environment variables to support reliable local development and dynamic data rendering from multiple platforms.

Android Developer Intern

- Developed *Bloom Book*, an Android application for plant growth tracking, implementing a responsive UI with data driven visualization features.
- Utilized Android Studio & Kotlin to develop interactive features and integrate a database for plant growth monitoring.
- Created detailed wireframes and interactive prototypes in Figma to refine user flows before implementation.

Academic & Research Experience

Qualcomm Institute – UCSD Serious Games Lab

September 2025 – June 2026

Software Engineer Intern (Game Systems & VR Development) – The Climate Games

San Diego, CA

- Built gameplay logic and visualization pipelines using Unreal Engine and C++ for real time interactive systems.
- Implemented a modular fire system to model ignition, propagation, extinguishing mechanics, and damage over time.
- Improved runtime performance by refactoring Blueprint-heavy gameplay systems into reusable components backed by native C++ logic.
- Developed and integrated Niagara based particle systems for fluid fire and smoke VFX, leveraging GPU particle simulation and parameter driven emitters for scalable environmental effects.

UCSD Cognitive Science Department

June 2024 – January 2026

UE Developer & Research Assistant, Systems Neuroscience Lab

San Diego, CA

- Supported Unreal Engine development for VR experiments investigating self-motion and trajectory perception.
- Architected immersive experimental environments with controlled motion parameters, stimulus timing systems, and data-logging pipelines for behavioral analysis.

Instructional Assistant & Programming Tutor

- **COGS 3 (Introduction to Computing):** Guided students through foundational web development concepts and assignments using HTML, CSS, and JavaScript.
- **COGS 18 (Introduction to Python):** Led weekly programming labs, supported students with Python assignments, proctored coding-based exams, and developed a Google Apps Script to identify students needing additional support.
- **COGS 187A (Usability & Information Architecture):** Mentored students in usability, information architecture, and interface design using Figma, including interactive systems for XR environments.
- **COGS 160 (Metabolic Health Analytics):** Assisted students with data processing, visualization, and time-series analysis in R for metabolic health applications.

UCSD Engineering Department - Global TIES

September 2024 – March 2025

UX Researcher, Cognovate Labs (Med-Tech Startup)

San Diego, CA

- Collaborated with EMTs and healthcare professionals to identify gaps in pre-hospital stroke assessment, addressing delays in critical medical decision making.
- Designed an EEG-based neural signal processing system with UX driven workflows to provide real time neurological insights for first responders.
- Researched electrocardiography (EKG) integration to enhance diagnostic accuracy and reduce latency in emergency stroke evaluation.

UCSD Computer Science & Engineering Department - ERSP

September 2023 – June 2024

Undergraduate ML Researcher - McAuley Lab

San Diego, CA

- Designed and implemented an LSTM-based sequence model for score to performance rendering, predicting expressive onset and offset timing from symbolic violin sheet music features using the Bach aligned MIDI-performance dataset.
- Engineered preprocessing pipelines to encode sheet music as structured tuples (onset, duration, pitch) and map them to ground-truth expressive timing targets for supervised learning.
- Developed PyTorch training workflows, implementing a custom loss function minimizing Euclidean distance between generated and real note boundaries across sequences.
- Benchmarked linear regression, MLP, and LSTM architectures to model temporal dependencies in expressive violin performance.
- Synthesized predicted expressive timing into playable MIDI outputs using MusPy for qualitative comparison against human performances.

Honors & Research Programs

Early Research Scholars Program (ERSP) – Certificate of Completion

June 2024

UCSD Computer Science & Engineering (CSE) Department

University of California San Diego

- Completed UCSD's competitive three quarter ERSP, receiving graduate level research training under CSE faculty.
- Collaborated with a small team to design, build, and present an original computer science research project.

Hispanic Scholarship Fund (HSF)

2024-2026

HSF Scholar

National Scholars Program

- Selected as an HSF Scholar, a nationally recognized designation awarded to high achieving students for academic excellence, leadership potential, and commitment to community impact.

UC San Diego McNair Scholars Program

November 2025 – August 2026

McNair Scholar - 3D Machine Learning Research

University of California San Diego

- Selected as a McNair Scholar based on academic merit and research potential, participating in a nationally recognized program supporting first-generation and underrepresented students pursuing doctoral studies.
- Designed a learning-based framework for adaptive 3D mesh optimization in XR systems, representing meshes as graph structures and applying Graph Neural Networks (GNNs) to predict vertex level importance for dynamic refinement and simplification.
- Developed a real time pipeline integrating geometric feature extraction (curvature, normals, visibility) with PyTorch Geometric and Unreal Engine to balance visual fidelity and computational efficiency under runtime constraints.

Education

University of California San Diego

September 2022 – June 2026

B.S. Cognitive Science (Human-Computer Interaction), Minor in Computer Science

La Jolla, San Diego, CA